CSTE Certification question dumps on Software Testing Manual Testing.

1. Premature release is defined as releasing the software into production under which of the following conditions EXCEPT?

1. **Test report is not prepared**
2. The test plan has not been completed
3. The requirements were implemented incorrectly
4. The test plan has not been completed.

2. Bottoms up integration testing has as its major advantages so that

1. major decision points are tested early
2. no drivers needs to be written
3. **no stubs needs to be written**
4. regression testing is not required

3. _____ provide direction, _____are the rules or measures by which the implemented policies are measured and the _____ are the means used to meet or comply with the standards

1. Standards, Policies , procedures
2. **Policies, standards, procedures**
3. Standards, procedures, policies
4. Policies, procedures, standards
5. Procedures, policies, standards.

4. Cost of non-confirmation includes

1. Training Survey etc.
2. Survey Studies
3. **Training Survey etc., Studies and Survey**
4. Rework

5.When should you stop testing

1. **When the quality goals established in the beginning of the projects are met**
2. when you run out of time
3. None of the listed options
4. when all the test cases are done

6. Some of the primary testing risks include EXCEPT

1. **Not Enough Training Lack of Test Competency**
2. **Lack of Customer and User involvement**
3. **Lack of management support**
4. **Lack of Test tools**

7. Given the following costs
User documentation $1000, Rework - 1500$, Design - 400$, Review of user documents - 800$, code review 1200$, The cost of quality is

1. $2400
2. $5000
3. **$3500**
4. $4000

8. Testing that may involve intentionally causing the system to fail so that recovery procedures can be tested is called

1. Stress testing
2. **Recovery testing**
3. Disaster testing

4. Security testing

9. Post Mortems Review is an Process Review (T/F) <u>True</u>

<u>10</u>. A Changes has been done to the design of an application, which of the following is required after this changes is implemented

1. Unit retesting
2. Subsystem retesting
3. None of the listed options
4. Design re verification
5. **All the listed options**

<u>11</u>. _____ are important elements of test objectives

1. Feasible and simple
2. Affordable and Flexible
3. Reasonable and understandable
4. **Reasonable and Adequate**

<u>(Note: Applicable, reasonable, adequate, feasible, affordable are all test objectives)</u>

12. Unit testing is conducted in the development environment (T/F). <u>True</u>

13. Test planning should begin

1. before Requirement collection
2. **<u>At the time of Requirement definition</u>**
3. after Requirement collection
4. after design

14. Summary project status report include information like

1. Resource, status, budget
2. **<u>Time, status, budget</u>**
3. Resource, budget, time
4. time, resource, budget

15. Following are the examples of satisfaction metrics EXCEPT

1. Ease of use
2. Acceptance criteria met
3. Customer subject assessment
4. **<u>Customer satisfaction</u>**

16. A white box testing technique that measures the number of or percentage of decision outcomes covered by the test cases designed is called

1. Consistent condition set
2. Basis path coverage
3. Modified decision coverage
4. **<u>Decision coverage</u>**

17. The functional testing that verifies the procedures to balance file function property is called

Compatability testing
<u>File integrity</u>
C**ompliance** testing
Integration testing

18. Activities asssociated with risk management are

Risk reduction methods
Risk analysis
Risk Migitation
Risk estimation

19. One of the most commonly identified weakness in software testing is
lack of time
lack of skilled resources
lack of metrics
lack of defects count

20. To develop a software test matrix follow these steps EXCEPT
Definite test plan
prepare the software test matrix
define test as required
define verification tests

21. Test coverage can be measured by EXCEPT _____
Traceability
Global data coverage
All of the listed options
User specified data coverage
Statement coverage

22. When one application feeds data to second application, the latter application is called as _____application
Regression
Stage containment
Downstream
Message Processing

23. The test tool that is used to test manual procedures in order to verify whether authorization procedures are followed is
Symbolic execution
Cause effect graphing
Inspections
Pareto analysis

24. Which of the following testing demonstrates key functional capabilities by testing a string of units that accomplish a specific function in the application?
Thread testing
Unit testing
Integration testing
Black box testing

25. Due to a change in design, the requirement for an already coded critical software component got modified in its entirely. The developer had to modify the code based on new requirement. Testing team is to choose the appropriate type of regression test to minimize the impact to the project schedule. Which type of regression test would you choose.
Full regression test
Unit regression test
Regional regression test
System Testing

26. Cost of confirmation includes

Training Survey etc.
Survey
Studies
Training Survey etc.,
Studies and Survey
Rework

27. The test schedule section includes which of the following except
Sequence of tests
Major test activities
The types of tests that must be conducted
Initial estimates for each activity

28. High priority in any defect management program should be
To improve the process
All the listed options
to prevent defects
to identify the best prevents on techniques and implement them

29. Testers insert additional code into a program to collect information about program behavoiur during program execution. This is called as _____
Instrumentation
Dynamic assertion
Inspection
Integration Testing

30. The inability of a system or components to perform its required functions when specified requirement is called
Bug
error
Failure
Defect

31. Which of the following statement about different risk type is correct?
It is not difficult to create test scenarios for high level risks
Strategic risks are the high level business risk faced by the software system
Tactical risk are subsets at a higher level of the strategic risks

32. Which among the following is NOT a way to define and measure test coverage?
User specified data coverage
Global data coverage
Stable/ Control Coverage
Modified Decision coverage

33. Which among the following is the correct sequence to test execution?
Identify test cases and test cycles, assign test scripts, set up test environment, review test cases
Set up test environment, identify test cases and test cycles, assign test scripts, review test results
Set up test environment, identify test case, review test result, assign test scripts
All the listed options

34. Following are examples of intentional act EXCEPT
False message may be inserted into the system
Communication protocol may fail to positively identify the transmitter or receiver of a message

Messages may be recorded and replayed into the system
If encryption (i.e use of codes) is used may be stolen

35. Which among the following is a suggested prioritization method to follow?

Critical, Major, Low
Major, Minor, High
Critical, Major, Minor, High
Critical, Low, High
Critical , Major, Minor

36. Without preparation of a test plan, at an earlier testing phase, tests may be _____

Economical
uneconomical and ineffective
None of the listed options
effective

37. Following are the valid defect types EXCEPT

Missing
Wrong
Extra
Change Request

38. The test technique to enter errorneous data to determine it cannot affect the integrity of the files is called _____

Security
Functional
Error handling
File integrity

39. Creating representative processing conditions using test transactions is the concept of _____

Test plan preparation
Test data preparation
All the listed options
Requirements Traceability Matrix

40. _____ is used to record data as it is gathered

Certification
Checksheet
Checkpoint
All the listed options

41. Test objectives _____

Enable the tester and project manager to gauge testing progress and success
Guide the development of test cases, procedures and test data
Enhance communication both within and outside of the project team by helping to define the scope of the testing effort.

42. One of the most effective method for estimating expected impact of risk is _____

Pareto Analysis
None of the listed options
Annual loss expectation formula
Risk analysis

43. A Static analysis technique that derives a symbolic expression for each program path is called _____

Structural testing
Statistical process control
None of the listed options
Symbolic execution

44. _____ in a test phase verifies the application being tested could correctly communicate with interrelated application system

Coupling
Functional testing
Unit testing
Regression testing

45. Which of the following statement is NOT correct?

90 % of all defects are caused by process problems
Reviews are generally greater than 65% efficient in finding defects
60% of software defects originate in the requirements phase of the project
Over 50% of the life cycle of software system is spent on requirements

46. Priority of defects should be given to on that caused data corruption, system crash and security violations (T/F). **True.**

47. Which among the following is a set of predefined action items used by test for a process to be complete and acceptable?

Formal analysis
Exit criteria
Force field analysis
Error guessing

48. When we talk about analysis performed by executing the program code, we are talking about _____

Force field analysis
Dynamic analysis
Formal analysis
None of the listed options

49. Following are the key concepts of the process engineering program EXCEPT

Absolute compliance to standard and procedures is required **(Process Engineering)**
Process is driven by testing projects (Not a Process Engineering)
Program is driven by management policies **(Process Engineering)**
Management provides an organizations structure for the workers to develop their own standards and procedures. **(Process Engineering)**

50. How do you track requirements in test case?

Requirement documents
Traceability Matrix
Test Plan
None of the above

Check for defects in process for the primary aim of correcting or establishing new process. This is

Both QA & QC
QC - (Correct Defect)
QA - (Establish process)
None of the listed options

52. Which of the following are the important elements of the control environment?

Ethical Values
Integrity Only
Integrity
Both Integrity & ethical values(Ans)

53. Testers implement the step-by-step method followed to ensure that standards are met. This method is called as _____

Coaching
Condition coverage
Completeness
None of the listed options

(Note: Ans is Procedure)

54. Which strategies are used to quickly find cause of defects?

Writing code to check the validity of system
Instrumenting code to trap state of system when anamalous condition occur
Analyze reported defects
All of the listed options

55. Expand the term COTS

Commerical Off The Shelf Software
Continuity of the software system
Continuity of the Shell software
Commercial of the system software

56. With thorough testing it is possible to remove defects from a program prior to delivery to customer (T/F). **True**

57. Which one of the following is the correct definition for procedure ?

The measure used to evaluate products and identify nonconformance. The basis upon which adherence to policies is measured.
Policies provide direction, standards are the rules of measures
managerial desired and intents concerning either process or products
The step-by-step method that ensure that standards are met.

58. _____ testing verifies all the performance criteria have been achieved when the system is placed into production.

System testing
Compliance testing
User acceptance testing
Security testing

59. To log incidents which of the following is used?

Test log
Test script
Defect tracking system

User report

60 Effectiveness is doing things right and efficiency is doing the right things (T/F). **False**

61. Testing _____ specified the type of test plan, recommended tools and techniques and type of documentation during testing phase.

Methodology
Process
Standards
Strategy

62. Following lead to improper use of technology EXCEPT

Improper skill
Detailed analysis of the technology
Early use of new software technology
Early use of new hardware technology

63. ALE Estimation stands for _____

Annual loss expectation
Actual loss expectation
Annual loss estimation
Actual loss estimation

64. Pareto voting is used in conjunction with

Cause and effect(Fishbone)
Vital few
80-20
Statistical process control

65. Compatibility testing for products involves all the following EXCEPT

Functional and non-functional compatibility
Certified and supported client environments
High and low level sanity testing
Client and Server side compatability

66. Guidelines to writing the test plans are EXCEPT

Calculate the planning effort
Finish early
Start early
Spend the time to do a complete test plan

67. Test team should investigate the following characteristics in order to evaluate the potential magnitude of the risk EXCEPT

Identify the components for the system
Asses the severity of potential failures
Evaluate contigency plans for this system and activities

Build the test plan

***68*. Which of the following is NOT a test factor?**

Portability
Compliance
Continuity of processing
Functionality
None of the listed options

69. The test design details which of the following

Sequence and increments of code delivery
State of the application at the start of testing
Availability of test equipment
The types of test that must be conducted

70. Following are the most effective way of communicating moral guidance EXCEPT

Written code of conduct
Lead by examples
CMM & ISO Standards
Verbal communication

71. Which one of the following is not a valid step to collect Benchmark data?

Analysis Phase
Integration Phase
Construction Phase
Planning Phase

72. Which of the following statement is correct?

The control environment is established by the highest levels of management and works upward through the organization

The control environment is established by the middle managentss and work downward through the organization

The control environment is established by the highest levels of management and works downward through the organization

The control environment is established by the tester and works downward through the organization

73. The primary goal of defect management is

to minimize the impact of defects
to prevent defects
to find defect as quickly as possible
All the listed options

73. Which among the following is the quantitative measure of the current level of performance

Affinity diagram
Pareto analysis

Benchmarking
Baseline

74.Regression test level best suites for which type of transaction?
Multiple transaction single terminal
Single transaction multiple terminal
Single transaction single terminal
Multiple transaction multiple terminal

***75*. During software acceptance, the testing technique that is relied upon for more accurate results(than any other testing technique listed below is_____**
Black box
White box
Incremental
Thread

76. Testing for the existence or effectiveness of programmed controls required using_____
Complex data
Invalid data
Redundant data
Valid data

77. Which of the following factor assures that data is processed in accordance with the intents of management for processing of transactions?
Service Levels
Authorization
Maintainability
Access control

78. The objective in Test design is to _____
Minimize test execution cost
defect as many defects as possible
minimize test maintenance costs
all the listed options

79. Severity is one which determines the order in which defects should be fixed (T/F). **False**

80. Steps involved in defect discovery are _____,_____and _____
Acknowledge, Find , Report
Find, Report, Acknowledge
Find, Acknowledge, Report

81. Tools for documenting defects as they are found and for tracking their status is
Defect tracking tools
Status Report

Test log

82. The best reason for using independent software testing teams is that_____

quality of software is given as much consideration as the project budget and timeline

software developers do not need to do any testing

testers do not get involved with the project until testing begins

stranger will test the software mercilessly

83. Which among process of analyzing and correcting syntactic, logic and other errors identified during testing?

Dynamic Analysis

Desk Checking

Debugging

Error guessing

84. Which of the following statement is correct about monitoring performance?

The risks to be organization and its products from using the process should be monitored

All the listed options

Management should review the process improvement program regularly

Careful analysis of results of testing should be performed.

85. Following are the key indicators to develop testers dashboards EXCEPT

Schedule status

Severity of recorded defect

Count of defects

Budget status

86. Testing on, above and below the edges of classes is called _____

Test contributions

Cause effect diagram

Equilance Class partitioning

Boundary value analysis

87.Which among the following are the major concerns testers have on entering the test execution?

Inadequate time or resources

All the listed options

Significant problem will not be uncovered during testing

Software not in testable mode

88. Which among the following are the techniques to prevent defects?

Training and eduction and methodology and standards

QA,Training and eduction, defensive design and code and finally methodology and standards

QA &QC

QA, Training and education, QC and methodology & standards

QA,QC, defensive design and code and finally methodology and standards

89. Which among the following is the best strategy to reduce impact if there is a problem?

Reduce scope of the system and also do not use latest unproven technology
Contingency plans
All the listed options
Disaster recovery plan, contingency plans

90. Test activities are recorded on a _____

Test log
Test plan
Defect tracking sheet
All the listed option

91. Which one of the following statements is correct?

Policies help in defining products
Policies are set by senior management
Policies are set by Test managers
Policies are guidelines

92. A testing method in which the test data are derived solely from the program structure is _____

Structural testing
Glass box testing
White box testing
All the listed options

93. Defects are measure in relation to _____

projects efforts
total number of defects
development team size
lines of code

94. The tool used to verify the operational system results are in compliance with organizations policies and procedures is called _____

Checklists and inspections
confirmation or examination
Fact fining
Tracing

95. To define test objectives testers need to do the following EXCEPT

Data bridges and filters
Validation process for incoming external data
Define each objective so that you can reference it by a number
Development of a model showing the internal and external dependency links among core business areas, processes and information systems

96. The tool used to examine the usability of test data for maintenance is called_____

Completion based analysis and mapping
Test data
Peer review
Inspections

97. Decision to stop test execution should be based upon _____

All the listed options
a % of coverage for each coverage category
rate of error detection falls below a specified threshold
successful use of specific test case design methodologies.

98. Which of the following is not a perspective of quality

Translucent
User Based
Transcendent
Product Based

99. Temptations causes employees to engage in the following improper acts EXCEPT

Weak internal audit function
High decentralization
Non existent or ineffective control
High Centralization

100. Following are guidelines adhered to tool usage to be more effective and efficient EXCEPT

The use of test tools should be incorporated into test processes so that the use of tools is mandatory, not optional

Tester should work together to build open source testing tools

Testers should have access to an individual in their organization , or the organization that developed the tool.

Testers should not be permitted to use tools for which they have not received formal training

101. The table of contents of a test plan might contain the following EXCEPT

Test objective
Test Scope
Assumptions
Test matrix

102. Which among the following is not a defect?

An extra function provided in the system
None of the listed options
A Requirement not implemented in the system
A Requirement implemented incorrectly

103. The concept of defensive code involves

adding a syntax checker in middle of code

None of the listed options

adding code to a program so that two parts of program must fail before a major problem occurs

adding throw() and catch()

104. Integration testing should begin once unit testing for the components to be integrated is complete (T/F) **True**

105. Which of the following is an informal analysis of the program source code yielding computer software ready for testing?

Code walkthrough

Code verification

Code inspection

Functional Review

106. Which of the following strategic issues needs to be addressed in successful software testing process?

Specify requirements in a quantifiable manner

All the listed options

Conduct formal technical review prior to testing

Consider using independent test terms

107. Which among the following is NOT a characteristic of good test case?

1) Clear to execute
2) It should be repeatable
3) It is not traceable to the requirements
4) **A good test case should be repeatable but not traceable to the requirements.**

108. Correct definition of vulnerability?

1. vulnerability is the potential loss to an organization as for example, the risk resulting from the misuse of its computer. this may involve unauthorized disclosure, unauthorized modification and/or loss of information resources, as well as the authorized but incorrect use of a computer. risk can be measured by performing risk analysis.

2.Vulnerability is something capable of exploiting vulnerability in the security of a computer system or application vulnerability include both hazards and events that can trigger flaws.

3. vulnerability is a design, implementation, or operations flaw that may be exploited by a threat, the flaws causes the computer system or application to operate in a fashion different from its published specifications and to result in destruction or misuse of equipment or data.

109. Yellow circle report signifies

1. serious problems

does not mean anything looks nice

no major problem

potential serious deviations

110. walter she wahart is known for his work on

1. control chart
2. histogram
3. **scatter plot diagrams**
4. run chart

Dump 2

1. Acceptance testing will be done by
   - **a. User**
   - b. Quality control
   - c. Quality assurance
   - d. Senior management

2. Main objective of configuration management
   - a. Maintain baseline for each version
   - b. Build software libraries
   - **c. Get the right change installed at the right time**

3. Customers will be get supported from these people for their problems
   - **a. Helpdesk**

4. Unit testing is a
   - **a. White box testing**
   - b. Black box testing
   - c. Stress testing

5. Before doing integration testing this testing must have been done
   - **a. Unit testing**
   - b. Systems testing
   - c. Stress testing

6. Cost of quality
   - **a. Appraisal, failure and preventive**

7. 4 types of testing
   - **a. System, unit, integration, acceptance**

8. There are 7 problem analysis tools. Which of the following is not one
   - a. Flow chart
   - b. Scatter plot
   - **c. Variance analysis**

11. Given the following costs:

User documentation - $ 1000 Rework - $ 1000
Design - $ 400 Review of User Documents - $ 1000
Code review - $ 1000

The cost of Quality is

a. $3000
b. $4000
c. $5000

12. Metric to measure operational reliability

a. MTBF
b. Defect density

13. The programmer writes a program step-by-step following this:

a. Procedure
b. Checklist
c. Process
d. Check sheet

14.Testing (including fixing the problem) is app. 50% of total product development cost

a. True
b. False

15. JIT is used only for inventory control by toyota

a. True
b. False

16. What is 6 sigma

a. 34 per 1,000
b. 34 per 1,00,000
c. 34 per 1,000,000
d. 34 per 10,000,000 (its 34 not 3.4 :-P)

17. Mean and median are

a. Central tendency
b. Variation

18. An example of deployment of a quality approach is:

a. The degree to which the approach embodies effective evaluation cycles
b. The appropriate and effective application to all product and service characteristics
c. The effectiveness of the use of tools, techniques, and methods
d. The contribution of outcomes and effects to quality improvement
e. The significance of improvement to the company's business

19. The concept of continuous improvement as applied to quality means

a. Employees will continue to get better
b. Processes will be improved by a lot of small improvements

c. Processes will be improved through a few large improvements
e. Improved technology will be added to the process, such as acquiring CASE tools
e. The functionality of the products will be enhanced

20. Quality is
a. Meeting requirements
b. Zero defects
c. Customer satisfaction
d. All of the above
a. a & c only

21. Systems are audited after implementation to determine whether or not the system met standards. This is an example of
a. Detective control
b. Quality control
c. Quality Assurance
d. Corrective control
e. None of the above

22. The primary responsibility of quality lies with
a. Senior management
b. Quality control

23. Random causes are assigned to normal variation
a. True
b. False

24. An operator verifies that all production jobs are run. This is
a. Quality control
b. Quality assurance
c. Production planning
d. Disaster recovery

25. The testing intended to show that previously correct software has not been adversely affected by changes is called:
a. Black-box testing
b. White box testing
c. Regression testing

26. The best phase that testing should begin is
a. After project starts
b. After requirements phase
c. After design phase
d. After coding

27. Assume you are a new quality assurance manager. You have observed a high failure rate in production due to defects in the computer programs. If you follow good quality practices, the approach you would take to reduce the incidence of failure is to:

a. Initiate code inspections to identify defects in code
b. Increase the type and extent of testing to remove defects before production
c. Classify and count the defects so that you can identify the defect of highest frequency and eliminate the root cause of the defect
d. Encourage the programmers to try harder to make fewer defects
e. Include individual defect rates in the performance appraisal system

28. The standards that have the greatest acceptance, and the highest compliance percentage, are normally those standards developed by

a. Senior management
b. The standards committee
c. A quality assurance group
d. The users of the standards
e. Internal auditors

29. Standard is:

a) **What**
b) When
c) Why
d) Where

30. Your project produces more than 100 defects

a. The project is useless
b. You cannot comment as you do not have much information

31. Your project produces 1000 defects per 150 function points. Now you have

a. Benchmark
b. Baseline
c. Size

32. The cost of correcting defects at the early stage of S/W development life cycle is less than correcting them at the later stage. **TRUE**/FALSE

33. Errors in a software are due to :

a. Faulty processes

34. Pareto chart is designed to rank items by frequency.

a) True b**) False**

35. One of most used QA tool is check sheet

a. True
b. False

36. How do you apply stress to the program

a. Larger volumes of transactions
b. Changes to transactions

37. The cost allocated for preventing the defects from occurring in right-first-time
a. Preventive
b. Appraisal
c. Failure
d. Direct

38. One attribute of risk is magnitude. What is other
a. User
b. Type
c. Frequency

39. After 5 years a project gave a profit of $ 50000. How will you measure the profit when the project was initially implemented
a. **Present value of money**
b. Return on investment
c. Asset depreciation

40. A place where data collected is recorded is
a. Check sheet
b. Histogram

41. Password should be changed
a. On regular basis
b. When you suspect that password is compromised
c. When you leave the place
d. All of the above
e. None of the above

42. On a random basis application systems are audited after implementation to determine whether or not the system met standards. This is an example of
a. Quality control
b. Detective control
c. Quality Assurance
d. Corrective control
e. None of the above

43. The effort required for locating and fixing an error in an operational program
a. Testability
b. **Maintainability**
c. Usability
d. Efficiency

44. The contributors to poor quality in an organization are
a. Lack of involvement by management
b. Lack of knowledge about quality
c. Time constraints

d. A and B

45. Definition of Integrity

a) Extent to which a program satisfies its specs and fulfils the user's mission and goals
b) Effort required for learning, operating, preparing input, interpreting output of a program
c) Effort required to couple one system with another
d) Extent to which access to software or data by unauthorized persons can be controlled
e) All of the above

46. Achieving quality is easy. True/**False**

47. "Failure to enforce standards" as a contributor to poor quality belongs to which category?

a) Lack of involvement by management
b) Lack of knowledge

48. To achieve quality (i.e., defect free products and services), we require

a) Close cooperation between management and staff
b) Commitment
c) An environment in which quality can flourish
d) All of the above

49. Quality is not a long-term strategy. True/**False**

50. PDCA cycle is developed by

a) Deming
b) Dr. Shewhart
c) Ishikawa
d) Harold S.Geneen

51. Quality Assurance (QA) and Quality Control (QC) are both necessary to make quality happen. Of the two QA is more important than QC

A. **True**
B. False

52. QC is a line function, performed by the same group that produces the goods

A. **True**
B. False

53. QA is a management function

A. **True**
B. False

54. QC is a related to the product

A. **True**
B. False

55. QA is related to the process producing the product

A. **True**

B. False

56. If you make the process consistent, then the product produced by the process will be consistent. This is main aim behind

A. Quality Control

B. **Quality Assurance**

C. Both of the above

D. Neither of the above

57. Check for defects in product with the primary aim of correcting defects. This is

A. **Quality Control**

B. Quality Assurance

C. Both of the above

D. Neither of the above

58. Check for defects in process for the primary aim of correcting or establishing new process. This is

A. Quality Control

B. **Quality Assurance**

C. Both of the above

D. Neither of the above

59. As per ISO protocols, all ISO standards should be confirmed, revised or with drawn in

a) Every 3 years

b) Every 5 years

c) Every 10 years

d) As required

Match the items in Tab A with Tab B.

Tab A	Tab B
1. ISO 9000	1. Guidelines for auditing the quality
2. ISO 9001	2. Guide to implement quality systems
3. ISO 9004	3. Describes quality models
4. ISO 10011	4. Introduction and guide to use other ISO 9000 standards

1-4, 2-3, 3-2, 4-1

61. The ISO 9001:2000 standards are based on **___8___** quality management principles.
As per ISO 9001 standards, verify the following statements.

A. The top management shall demonstrate its commitment to meet customer requirements for Their products and services (True / False).

B. The organization should determine and provide the resources needed to establish and improve quality management system (True / False).

C. The organization should ensure that all processes are operated under controlled conditions and produce outputs that are consistent with organization's policy and objectives (True / False).

D. The organization should define and implement measurement, analysis and improvement processes as a way of demonstrating that the product/service conforms to specified requirement. (True / False)

62. What are the two types of reviews defined in ISO/IEC 12207
Ans. Joint technical review, Joint management review

63. Management commitment is the single most important requirement for successful implementation of ---------- (answer - Quality Management)

64. Having management commitment does not guarantee quality management success but improves odds for successful implementation. (True/False)

65. Top down implementation model(waterfall effect) is most successful model for implementing quality management. (True/false)

66. Committing to quality management means
a. Understand the concept of quality management.
b. Adopt behavior to show quality management.
c. Accept the need to change the perspective leadership.
d. Provide funds for training.
e. All of the above

67. Which is the first step in establishing the trust among employees in implementation of quality
a) Actively listening down
b) Talking down
c) A) or B)
d) None of the above

68. Understanding comes after behavior change (True/False)

69. Toyota Production System is also called as
a. Just In Time
b. Join In Time
c. Single Minute Exchange of Dies
d. None of the above

70. Visiting other companies and understanding its quality process reduces the lead-time for identifying the quality process to be implemented in an organization. True/False

71. Which of the following is the most difficult of the decision making processes?
a) Authority
b) Voting
c) Avoidance
d) Consensus

72. The objective of constructive criticism is
a. to criticize some one on their bad works
b. to improve his performance

c. duty of the project manager

73. Whether it is a must for the listener to listen to all parts of speech? YES

74. Which of the following is discriminating listening
a) Directed at selecting specific pieces of information, not the entire communication.
b) Complete message with minimal distortion
c) To put yourself in the speakers situation
d) To analyze what the speaker has said.
e) None of the above.

74. Which of the following is Therapeutic listening
a) Complete message with minimal distortion
b) Directed at selecting specific pieces of information, not the entire communication.
c) To put youself in the speakers situation
d) To analyse what the speaker has said.
e) None of the above.

75. Which of the following help attending to the speaker
a) Maintain eye contact
b) Provide continuous feedback
c) Periodically restate what was said
d) Concentrate exclusively on the speakers communication
e) All the above

76. Which one below is not an attribute of executive management
a) Purposeful
b) Competent
c) Analytical
d) Decisive
e) Meaningful

77. Which attributes were included in establishing status when equated to IT environment.
a) leadership
b) technical skill
c) years of experience
d) all the above.

78. Quality promotes turning problems into opportunities (True / False)

79. Customer will give up if complaints are not resolved within
(a) 4 Days
(b) 4 Months
(c) 4 Seconds
(d) 4 Minutes

80. Following is part of complaint resolution process
(a) Get on your customer's wavelength
(b) Follow up with the customer
(c) Establish and initiate an action program
(d) All the above

81. Writing a successful report requires a clear understanding of both _____ and _____
Ans: Report objective and desired action.

82. A good report has no more than _____ and _____
Ans: Three objectives and three actions.

83. QA report is designed to
(a) Convey information
(b) Change behavior
(c) Trace problem
(d) None of the above

84. The scope of a IT Quality function includes
a. Supporting implementation of quality policy
b. Reducing producer and customer gap
c. All of the above
d. None of the above

85. When the quality processes are mature with defined, improving work processes the focus of the quality function is on
a. Optimization
b. Product quality
c. Process deployment and compliance
d. All of the above

86. Why are product inspections needed?
a. To find out the best worker
b. To find the worst worker
c. To assist the worker in improving quality
d. For objectively appraising workers
e. All of the above

87. Which is the least important skill for a quality manager in the below list?
a. Verbal communication
b. Project management
c. Programming
d. Systems knowledge
e. Knowledge of operations
f. Written communication

88. Which is the best positioning of a quality manager?

a) Reporting to manager of systems/programming
b) Reporting outside the IT function
c) Reporting to manager of computer operations
d) Reporting to IT manager

89. The quality function is normally the weakest when reporting to _ _ _ _ _
a) Manager of systems/programming
b) Outside of IT function
c) Manager of computer operations

90. Based on research from the Quality Assurance Institute the percent of quality groups reporting outside the IT function is 15%. YES /NO

90. The size of the quality function ideally will vary with the size of the organization (True/False)

91. Obtaining and sustaining support for quality is an ongoing activity - TRUE/FALSE.

92. Which are the following tasks is not helpful in building support of quality
a. Teach a quality vocabulary
b. Issue a quality newsletter
c. Staffing
d. Meet daily on quality

93. In any organization QA Analyst position will be
a. Part time staff position
b. Full time staff position
c. Not required at all

94. QA Analysts have to be trained quality functions YES

95. Which are following disciplines is helpful in performing the quality function
A. Auditing
B. Individual engineering
C. SPC
D. All the above.

96. Which of the below tool help people to determine whether a process is in control or not

a. Run Chart
b. Control Chart
c. Pareto Chart
d. U Chart

97. Which are following tools belong to Quality Tool Box

a. Brainstorming
b. Affinity Diagram
c. X-mR chart
d. Pareto Chart

98. What is the most commonly used / Powerful statistical tool?
Checklist

99. What is a simple list of questions that enables people to follow and verify processes?
Checklist

100. _____ is a set of tools, which assists in defining, controlling and improving quality.
Quality Toolbox

101. List the tools commonly used by QA Analysts

- Brainstorming
- Flow charts
- Cause-and-effect diagram
- Histograms
- Pareto Charts
- Control charts
- Scatter diagrams

102. List few plans, a quality manager can use for driving and continually maturing the qualityenvironment

- Stay abreast with new quality practices
- Study the change in criteria for different quality models
- Frequently review the quality benchmark
- Obtain and Maintain quality related certifications

103. How the role of QA changes as the quality function matures in an organization?
Quality control to Quality Assurance to Quality consulting

104. Suggest another name for Independent Testing Department
Quality Control Department

105. Quality practitioners working in Quality control department are called _____.
Inspectors or Testers

106. What are the major activities in the 'Initial' phase of quality management maturation?

- Perform Quality control activity
- QA activities focuses on process definition- but not fully defines and follows the process

107. _____ is the prerequisite to improved quality and productivity.
Consistency

108. What are the major activities in the 'Intermediate' phase of quality management maturity?

- Quality control becomes shared responsibility of the customer, worker and QA analyst
- Process definition and improvement are emphasized and performed
- Quality function and QA analyst acts as a quality consultant

109. How does MBNQA program define an organization which moves towards optimization?
World-Class

110. What is the main identity of a World-class Organization?
Yield high customer satisfaction at minimum cost.

111. What are the major activities in the 'Final' phase of quality management maturity?

- Workers are assigned responsibilities for process definition, measurement and improvement
- Role of QA analysts changes to performing Quality consulting to management and employees in promoting and implementing quality initiatives

112. What are the two major drivers that change the role of the QA analyst

- Management Philosophy used in the IT group
- Personal belief system of managers

113. In the 'Initial' phase of quality function maturation, what type of management exists?
Authoritarian

114. When quality management philosophy matures, the organization moves from hierarchical structure to _____.
Teams (that are organized and empowered to define, measure and improve processes.)
Flowcharts may be a:

a) Simple High level process flow
b) A detailed task flow
c) Any mix of a and b
d) All of the above

116. Flowcharts are useful to obtain knowledge of a **process for improvement.** It helps to document a process so that**, inefficiencies** and **redundancies** can be identified and reduced.

117. A process map is a more detailed flowchart that depicts:

a) Processes
b) Relationship between Processes
c) Process Owners
d) All the above

118. Flow Charts should reference the following:

a) Process owners, suppliers and Customers
b) Key deliverables
c) Decision points
d) Interfaces
e) Task and Task Sequences
f) Policies, Standards, procedures and Tools
e) All of the above

119. The process of measuring how well a company's products, services and practices measure against others is known as **benchmarking.**

120. Benchmarking can be done against partners who could be

a) Internal Units
b) Competitors
c) "Best in class" or "best in breed" within any industry
d) Any of the above

121. The objective of benchmarking is to identify performance gap and to determine a realistic improvement goal.

122. List the three types of benchmarking

a) Process Benchmarking
b) Product Benchmarking
c) Performance Benchmarking

123. Matrix is a

a) Problem Solving Technique
b) Problem Identifying Technique
c) Problem Cause Identifying Technique
d) Ranking Technique

124. Matrix is also known as _____ and _____
Check sheet, matrix diagram

125. The two common types of Matrixes are

a) L-Type and X-Type
b) X-Type and V-Type
c) L-Type and T-Type
d) T-Type and X-Type

126. L-Type matrix compares

a) 2 sets of items
b) 3 sets of items
c) none of the above

127. The Principal Developer of QFD (Quality Function Deployment) is

a) Dr. W. Edwards Deming
b) Dr. Joseph Juran
c) William Scherkenbach
d) Dr. Yoji Akao
e) Dr. Harold S. Geneen
f) None of the above

128. Quality System is an Organized approach to quality with _____, _____ and_____.

Tools, techniques and methods

129. Comprehensive Quality Deployment includes _____ , _____ , _____ and_____
Quality, technology, cost/schedule and reliability

130. Scatter plots are used for
a. Problem solving and understanding
b. Cause and effect relationships
c. Test influence of one variable over the other.
d. All of the above

131. Which of the following use Scatter plots
a. Defect level vs.Skills
b. Cost vs time
c. change response
d. Defect level vs Complexity
e. All of the above

132. The following are not types of scatter plots
a. Positive
b. Negative
c. Zero
d. Non zero

133. Which of the following are graphical presentation tools.
a. Table
b. Pie chart
c. Scatter plot
d. Bar chart
e. None of the above

134. Which of the following charts show direction of events
a. Bar chart
b. Run chart
c. Line chart
d. Table

135. Line charts can be used for the following:
a. Like units
b. Like periods
c. Like variables
d. Related or fixed variables.

138. The following are 2D charts:
a. Bar charts
b. Line chart
c. Run chart
d. Control chart

139. Data precision is not highly required for

a. Line chart
b. Control chart
c. Bar chart
d. pie chart

140. Which of the following chart represents data in percentage

a. stem and leaf chart
b. Bar chart
c. Pie chart
d. Line chart

141. Stem and leaf chart is a variation of

a. Line chart
b. Pie chart
c. Bar chart
d. Control chart
e. None of the above

142. It is practical to use stem and leaf charts when absolute no of values is

a. High(More than 100)
b. Low (Usually 100 or less)
c. Zero
d. None of the above

143. Testing performed by organizations other than the developer is

(a) Unit testing
(b) Integration testing
(c) System testing
(d) Acceptance testing
(e) Independent testing

144. "Validate that the functions meet stated requirements" is

(a) Unit testing
(b) Integration testing
(c) System testing
(d) Acceptance testing
(e) Independent testing

145. "Validate that all units work together according to the system design" is

(a) Unit testing
(b) Integration testing
(c) System testing
(d) Acceptance testing
(e) Independent testing

146. "Validate that system meets the needs of the business" is

(a) Unit testing
(b) Integration testing
(c) System testing

(d) Acceptance testing
(e) Independent testing

147. "Incremental testing" means
(a) White Box testing
(b) Black Box testing
(c) Top-down testing
(d) Regression testing
(e) Independent testing

148. "Data driven testing may be called as
(a) White Box testing
(b) Black Box testing
(c) Regression testing
(d)Independent testing

149. Logic driven testing may be called as
a. White Box testing
b. Black Box testing
c. Top-down testing
d. Regression testing
e. Independent testing

150. Verification should be performed for
(a) requirements
(b) design
(c) code construction
(d) operation / maintenance
(e) all of the above

151. Validation is mostly used to determine the _____ of the final software / program
(a) Correctness
(b) Completeness
(c) Consistency
(d) Quality

152. Verification is mostly used to determine the <u>effectiveness / completeness</u> of the software

153. For small programs, verification is the process of determining the correspondence between a program and its specifications
True / **False**

154. As part of configuration management, small changes need not be documented and controlled.
a. True.
b. **False.**

155. Someone independent of the team that designed the change should be involved in testing the final revised program.''
a. **True.**

b. False.

156. The primary goal of defect management.

a. Defect correction.

b. **Defect prevention.**

c. Defect reduction.

d. All of the above.

157. The defect management, unlike the entire s/w development process need not be risk driven.

a. True.

b. **False.**

158. The general principles of recording management includes,

a. **Preventing defects.**

b. Capture and analysis of information should be automated.

c. Should focus on the person who caused the defect.

d. Defect information should be used to improve process.

159. The main purpose of recording defects are

a. To ensure the defect is corrected.

b. To gather statistics used to develop defect expectations in future applications.

c. To go for continuous improvement.

d. **All of the above.**

160. Severity of defects should be assigned subjectively, whereas priority should be assigned objectively.

a. True.

b. **False.**

161. How many steps will be there in simple Defect Tracking Process (DTP).

a) 2
b) 3
c) 4
d) Just one

162. From the following what are the steps in DTP

a) Execute test and log any discrepancies
b) Draw Fish bone diagram
c) Assign defect to a developer
d) Do causal analysis

163. What are the recommended levels to 'Prioritize the correction'

a) Critical
b) Major
c) Minor
d) Oversight or inadequate self review

164. Test reports issued periodically include ---------- by severity or priority

a) Summary of all the defects found

b) Summary of all closed defects
c) Summary of all open defects
d) None of the above

165. Test reports are issued to
a) Team alone
b) Team and Management
c) Management alone
d) None of the above

166. Test reports are issued
a) Periodically throughout the testing process
b) At the end of testing process
c) Once in a month
d) Once in the lifetime of a project

167. Defects can also be used for Process Improvement
True / False

168. Today many organizations use defects for Process Improvement
True / **False**

169. Based on research team findings, using defects for Process Improvement should include the following
a) Go back to the person who did coding
b) Go back to the process that originated the defect to understand what caused the defect
c) Go to the other project to know how they dealt with that problem
d) Go back to the verification and validation process, which should have caught them earlier

170. Human factors dimension alone can have a very large impact on the effectiveness of the review process
True/False
1. Standard states

a) what must happen to meet the intent of the policy

b) how to meet the requirements

c) both A and B

d) None of the above

172. Standards convert intentions into specific rules (true / false)

173. Standards must be

a) measurable

b) attainable

c) necessary

d) nice to have

e) all the above

174. Which of the following determines standards

a) customer requirements

b) internal requirements

c) cost and schedule requirements

d) all the above

175. Which of the following is true

a) procedures are not always required and unless critical, should be minimised

b) skill set and knowledge requirements are prerequisites for performing the procedure

c) if the procedures are followed ALWAYS standards will be automatically met

d) All of the above

e) none of the above

176. Controls should be designed based on the criticality of the process and what needs to be checked(True / false)

177. Quality control procedures are

a) preventive costs

b) appraisal costs

c) failure costs

d) none of the above

178. Measurement provides quantitative feedback to an organization about whether it is achieving its goals and moving towards its results (**true** / false)

179. The purpose of process improvement is

a) to change the process

b) to reduce the frequency of defects including process ineffectiveness

c) both a and b

d) none of the above

180. Process improvement uses facts(measurement results) to identify the root causes of problems.

(**true** / false)

181. The long range objective for process improvement is to

a) eliminate the need for quality assurance activities

b) eliminate the need for quality control activities

c) both a and b

d) none of the above

182. IF THE CONTROL LIMITS ARE WITHIN THE SPECIFICATION LIMITS, THE PROCESS IS CAPABLE OF SATISFYING THE CUSTOMER. (TRUE/FALSE**)**

183. Does the process improvement plan need approval? If yes, who approves it ?

Yes, Senior management

184. MEASURE IS THE BASIC BUILDING BLOCK FOR MEASUREMENT PROGRAM.

185. Metrics is a derived unit of measurement

186. What are the two types of measurement?

Objective and Subjective Measurement

Objective measurement uses hard data that can be obtained by counting, weighing etc.

Subjective measurement is normally observed or perceived

187. What is Nominal data?

Nominal data cannot be subjected to arithmetic operations of any type and the values cannot be ranked in any natural order.

188. What is Ordinal data?

Ordinal data can be ranked, but differences or ratios between values are not meaningful.

189. What is interval data?

Interval data can be ranked and can exhibit meaningful differences between values

190. Give an example for interval data?

T.J. McCabe's complexity metric is an example for interval data.

191. What is Ratio Data?

Ratio data has absolute zero and meaningful ratios can be calculated.

192. For what are the measurement dashboards or key indicators used for?

Measurement dashboards or key indicators are used to monitor progress and to initiate change.

193. Before approving for use, what are the tests that the measures and metrics have to undergo?

1) Reliability – Refers to the consistency of measurement

2) Validity – indicates the degree to which a measure actually measures

3) Ease of Use and Simplicity

4) Timeliness – Refers whether the information can be reported in sufficient time to impact the decisions needed to manage effectively

5) Calibration - indicates the modification of a measurement so it becomes more valid

194. How to know a measurement is good?

The measures and the resulting metrics should be

1) Simple and precisely definable

2) Objective

3) Easily Obtainable at reasonable cost

4) Valid, measuring what they are intended to measure

5) Robust, being relatively insensitive to intuitively small changes in the process or product

195. Give 2 examples of metrics for complexity measurement s?

a) Cyclomatic complexity

b) Knots

196. A knot is defined as a necessary crossing of directional lines in the graph.

197. The quality characteristics for software that have received considerable attention are correctness, reliability and **maintenance.**

198. Common causes are causes of variation that can be controlled by improving work processes.

199. Special causes are causes of variation that must be controlled outside the process.

200. Statistical tools are the only methods available to objectively quantify variation.

201. How many Components are in Risk and name all of those Components?

Three components

- The event that could occur – the risk
- The probability that the event will occur – the likelihood
- The impact or consequence of the event if it occurs – the penalty

202. Risks can be categorized as one of the following:

- Technical
- Programmatic or performance
- Supportability or environment
- Cost
- Schedule

203. How many Characteristics does Risk has and what are they?

Five Characteristics

- Situational
- Time based
- Interdependent
- Magnitude dependent
- Value based

204. PMBOK stands for?

Project Management Body of Knowledge

205. How many processes are there to address risk management as per PMBOK?

Four processes

- Risk Identification
- Risk Qualification
- Risk Response Development
- Risk Response Control

206. External risks can be controlled or influenced by the project team ---------- TRUE/**FALSE**

207. Which of the following document will be required as a source of input when considering a list of possible risks?

a) Product documentation b) Project documentation c) Historical Information **d) All of the above**

208. In a traditional waterfall methodology, most risk management activity occurs close to milestone. **True**/false

209. In a spiral development model the risk management activity falls in the explicit risk management portion. **True**/false

210. Risk management is an independent auditing process. True/**False**

211. Risk should be the sole responsibility of project manager. True/**False**

212. Within many software development organizations risk management remains <u>ad hoc</u> and <u>incomplete.</u>

213. Testing strategies is included in short term risk management. True/False

214. Who should be involved in determining risk management

a. Customer
b. Management
c. Development team
d. **All of the above**
e. None of the above

215. Mitigating the risk by

a. Minimizing the probability of occurrence
b. Minimizing the value of impact
c. Deflecting the risk elsewhere
d. All of the above
e. None of the above

216. A measurement program is defined as the ----------

Ans: Entire set of activities that occur around quantitative data.

217. Mature organization use quantitative data to determine

a. **Requirements are complete**
b. **To prevent problems**
c. Budget
d. Schedule and project status
e. All of the above

218. Immature organization use quantitative data to determine

a. Requirements are complete
b. To prevent problems
c. **Budget**
d. **Schedule and project status**
e. All of the above

219. What are four major uses of Measurement?'

Ans: 1.Manage and control the process

2. Manage and control the product (review and testing)

3. Improve the process (quantitative data gathered during process execution can identify the weaknesses and therefore opportunities for improvement)

4. Manage the risks

220. Same software development process may be followed to produce two different applications. - True/False

221. What is risk?

Ans: Risk is a chance for something to go wrong.

222. How do we manage control and product?

Ans: Measure **Quality** by no.of defects, Mean time to failure (MTTF) Mean time to repair (MTTR)

Measure **Performance** by technical performance, measures specified by customer and management.

223. How do we improve the process?

Ans: Identify the current performance; find how cost-efficient the process is. By measuring Time and effort.

224. How do we manage risks?

Ans: Identify the risks, its frequency, probability of risks occurring and impact of each risk if it occurs derive risk management plan.

225. What are the four phases in installing a measurement program?

Ans: 1.build the measurement base

2. Manage towards results

3. Manage by process

4. Management by fact

226. What are the key points in first phase of implementing a measuring program?

Ans:1. Define the objectives for the measurement program

2. Create an environment receptive to measurement (establish SLA between IT and users, establish quality measurement environment)

3. Define the measurement hierarchy

Three levels of quantitative data are mapped to 3-level organization tier.

a. **Metrics** – used by **Line management** such as variance between actual and budgeted

cost, user satisfaction, defects rates per LOC.

b. **Measures** – **Staff management** uses basic measures such as product size, cycle time or defect count.

c. **Strategic results** – **Senior management** uses strategic results dashboard to manage the IT function and track to mission, vision and goals.

227. What do we mean by manage towards results?

Ans: a. Identify goals for desired business results

b. Means for measuring each business result

228. What are the five steps in manage towards results phase?

Ans: a. Identify desired business results. If each has subject, action, target value and time frame, as it is much easier to identify measure for each.

b. Identify current baseline

c. Select a measure or metric for each result. Should be only 3 to 5 but not greater than 7

d. Consider tradeoff between the results. For example consider the desired result **Complete on time** might affect **Easy-to-read documentation**.

e. Based on baseline and desired result, determine a goal for each result metric. (Bench marking tool is best for this).

229. Quality is an attribute of

a) Product

b) Process

c) Standard

d) Policy

230. What does "IEC" expands to (E.g. As used with ISO/IEC 12207)?

a) International Electronics Committee

b) International Engineers Confederation

c) International Electrotechnical Commission

d) Internal Electronics Association

231. Which of the following defines the "SPICE"?

a) SEI CMM

b) ISO/EIA 12207

c) ISO 9001:2000

d) ISO/IEC TR 15504

232. The two components of the “Manage By Fact” process are

a) Meeting the desired results

b) Managing to meet the customer requirements on fact gathering

c) Managing the process to drive the results

d) A & C

e) B & C

233. Which of the following is not a characteristic of a ‘Service Level Agreement’?

a) It defines the obligations of the services/product provider towards the customer

b) It has terms for maintenance of the product/service

c) It does not mention the obligation of the customer

d) None of the above

234. Function point analysis involves

a) Amount of information processing function

b) System characteristics

c) Degree of influence

d) All of the above

e) None of the above

235. Why does it help to visit other companies to identify effective Quality practices?

a) Reuse organizational learning

b) Continually benchmark

c) Interact with other Quality professionals

d) All of the above

e) None of the above

236. Which step of the listening process requires to give continuous feedback

a) Hearing the speaker

b) Attending to the speaker

c) Understanding the speaker

237. Any additional feature required by the customer, which is not mentioned in the requirements is a(n) _____ from the producers' point of view:

a) enhancement

b) defect

c) modification

d) None of the above

238. What are the three Cs in the today's IT workplace (of QA profession within IT)

a) Completeness, Change and Conformance

b) Change, Conformance and Configuration

c) Change, Complexity and Competition

d) None of the above

239. Audit plan is prepared during the definition phase.

a) True

b) False

240. Who created the theory of FIRO

a) James Baehler

b) W.C. Shutz

c) Taichi Ohno

d) Dr. Ishikawa

241. Given the following costs:

Requirements analysis - $ 1000 Rework - $ 700

System Testing - $ 500 Phase Reviews - $ 650
Code review - $ 500 Review of User Documents - $ 600
Continuing Edn. Programs. - $ 800 Design - $ 400

Client Entertainment - $ 800

The cost of Quality is

a) $ 4500

b) $ 1100

c) $ 1600

d) $ 3750

e) $ 3500

242. In an organization, the process is stabilized and documentation exists. What is the rating for this level?

a) level 1

b) **level 2**

c) level 3

d) level 4

e) level 5

243. Malcolm Balridge who was a defense secretary invented MBNQA

a) True

b) **False**

244. Industry accepted definitions of quality are "conformance to requirements" (from Philip Crosby) and "fit for use" (from Dr. Joseph Juran and Dr. W. Edwards Deming). These two definitions are contradictory.

a) True

b) False

245. Most Information Technology (IT) groups have the quality gaps between: ----------.

a) Producer gap

b) Supplier gap

c) Customer gap

d) Provider gap

e) A & C

f) A & B

246. A major premise of a quality management environment is an emphasis on continuous improvement. The approach to continuous improvement is best illustrated using the ----------, which was developed in the 1930s by Dr. Shewhart of the Bell System.

a) System development life cycle

b) PDCA Cycle

c) Quality Improvement Cycle

d) Product Life Cycle

247. The structure and application of controls must be clear to all parties concerned. In this respect, two levels of control exist within an environment:

a) Management controls.

b) Application controls

c) System Controls

d) Quality controls

e) A & B

f) B & C

248. Defects are recorded _____

a) to report status of the application

b) to gather statistics used to develop defect expectations in future applications

c) to improve the software development process.

d) to ensure the defect is corrected

e) All of the above

249. The system design SDLC phase is immediately followed by

a) Initiation

b) Definition

c) Evaluation

d) Programming and training

e) Operation

250. Resource planning, audit planning, estimation, scheduling are the some of the tasks carried out in

a) Initiation phase

b) System design phase

c) Definition Phase

d) Evaluation phase

251. Internal control encompasses the plan of organization and all of the methods and measures adopted within a business to:

a) Safeguard its assets

b) Promote operational efficiency

c) Check the accuracy and reliability of accounting data

d) Encourage adherence to prescribed policies

e) All of the above

f) None of the above

252. As part of management control, the division of responsibilities provide

a) efficiencies derived from specialization

b) cross-check that promote accuracy without duplication or wasted effort

c) safeguarding of assets

d) enhances the effectiveness of a management control system

e) All of the above

f) None of the above

251. System reviews and software testing are examples of _____.

a) Quality Control

b) Quality Assurance

c) Quality Audits

d) None of the above

252. The _____ is the process used to verify and validate the system structurally and functionally.

a) Tester's workbench

b) Developer's workbench

c) Owner's workbench

d) Manager's workbench

253. _____ testing is done without executing the code.

a) Regression

b) Unit

c) System

d) Static

254. Identification of defects within the stage or phase of the project where they originate, rather than in later test stages is referred to as _____.

a) Stage containment

b) Early catch

c) Defect Prevention

d) Defect containment

255. Informal, unstructured reviews of the program source code against specifications and standards to find defects and verify coding techniques are called _____ .

a) Inspection

b) Buddy reviews

c) Code walkthrough

d) Peer Review

256. Which of the following is not a white box testing technique

a) Statement coverage

b) Equivalence partitioning

c) decision/condition coverage

d) multiple condition coverage

257. _____ is the way of Incremental testing which begins testing from top of the module hierarchy and works down to the bottom using interim stubs to simulate lower interfacing modules or programs.

a) Bottom-up

b) Top-down

c) Lateral

d) Block-by-block

258. Which of the following task is not performed by V&V management

a) Create the software V&V plan

b) Conduct Management Review of V & V

c) Support Management and Technical Reviews

d) Interface with Organizational and Supporting Processes

e) Conduct In-process reviews

259. Awareness Trainings should be limited to

a) 1/2 hour

b) 1 hour

c) 2 hour

d) 3 hour

e) 3-4 hour

260. The ultimate goal of JIT is supply each process –

a) the exactly required item

b) in exactly required quantity

c) at exactly required time

d) All three

261. Pull system means –

a) Quickly changing work capabilities

b) Reducing errors, defects, waste

c) Data is moved forward when it is needed by next operation.

262. Mercury's test director is a tool to support _____ tracking process.

a) Requirements

b) Defect

c) Change Requests

d) Leave taken by testers

263. Probability of a software failure ----------

a) Defect rate

b) Reliability

c) Correctness

d) Testability

264. Who is responsible for risk management?

a) Team members

b) Project Leader

c) Project Manager

d) Program Manager

265. Major uses of quantitative data

a) Manage and control the process

b) Manage and control the product

c) Improve the product

d) Manage the risks

e) All of the above

266. Function Points

a) measure of the system size

b) can be used to compare different kinds of application systems

c) together with past data can be used to estimate future effort

d) all of the above

e) none of the above

267. Workbenches are also called _____

a) Phases

b) Steps

c) Tasks

d) All of the above

e) None of the above

268. Five components of a process represent the vocabulary of process management –

a) Policy, Procedures, Training, Input and Output/deliverables.

b) Vision, Mission, Objectives, Goals, and Process

c) Policy, Standards, Inputs, Procedures and Output/deliverables.

d) Policy, Guidelines, Procedures, Standards and Milestones

269. Business Control model includes three general areas of control –

a) Management processes Control, Work processes Control and Check Processes Control.

b) Statistical Process Control, Statistical Quality Control and Statistical Product Control

c) Process Control, Product Control and Environment Control

d) Quality Control, Management Control and Customer Control

270. A standard must be _____

a) Smart, Measurable, and Time-bound

b) Measurable, Achievable and Clear

c) Approved, Available and Attainable

d) Measurable, Attainable and Critical

271. Most important requirement for successful implementation of quality management is:

a) Employee Awareness and Commitment,

b) Succesful Deployment of Qulaity Function,

c) **Management Commitment,**

d) Commitments from people involved in Quality Group

271. Slowest group and weakest link in most quality management efforts in an organization is:

a) Executive Management

b) Middle Management

c) Line Management

d) Other Employees

271. The management effort to create a new culture, which encourages and accepts changes and allows all people to work together to maximize their contributions to quality management is a

a) Mission

a) **Vision**

b) Goal

c) Policy

d) Objective

272 What is the number one skill for a QA ANALYST?

a) Verbal Communication

b) Written Communication

c) Listening

d) Leadership

273 Each Key Process Area is organized in to 5 sections called ..... .And they are ..... ,.....,....., ..., ..., ...

(27) Common Features. Figure 3-2 in pg 3-7

commitment to perform, ability to perform, activities performed, directing implementation, verifying implementation.

274. What is NIST?

National Institute of Science and Technology

National Institute of Standards and Technology.

National Institute for Software Technology

National Institute for Software and Technology

275. The ISO 9000:1994 family of Standards contained..... (number)standards and documents.

18, 24, **20**, 22

276. Which are the four primary standards of IS0 9000?

ISO 9000, ISO 9001, ISO 9004, ISO 10010

ISO 9000, ISO 9001, ISO 9006, ISO 10011

ISO 9000, ISO 9001, ISO 9004, ISO 10011
ISO 9000, ISO 9001, ISO 9004, ISO 10054

277. What are the 4 views of Software Development?

Requirements View, Design View, Coding View, Testing View

Manager View, Developer View, SQA View, Customer View

Contract View, Operating View, Engineering View, Supporting View

All of the above

None of the above

278. List out the six phases that are involved in a SDLC?

A) Initiation - Phase I

B) Definition - Phase II

C) System Design - Phase III

D) Programming and training - Phase IV

E) Evaluation and Acceptance - Phase V

F) Installation and Operation - Phase VI

279. Producer Views Quality is meeting ----------.

A) Whether the Product or Service Does What the Customer Needs

B) Product Has Zero Defects

C) Producer's Requirements or Specifications

D) Conformance to Requirements

280. As a Quality Assurance professional, you will be doing

a) Reviews, Walk-throughs , and inspections

b) Unit Testing

c) Statistical Process Control

d) Identify the weak areas in a SDLC, take corrective action, so as to ensure it does not occur again

281. "Benchmarking" is

a) A measure of the current position of your project/product

b) Measure of the present position of your project/product compared to a previous state

c) A measure of the present position of your project/product compared against a future (estimated) state

d) A measure of your product compared to an existing industry standard.

282. The Management Cycle, As Per The ----------, comprises Of Plan, Do, Check And Act.

A) QAI's Quality Improvement Model

B) ISO 9000

C) SEI Model

283. Which, of the four given below, is not one of Dr. Edward Deming's 14 Quality Principles?

a. Give adequate training

b. **Have a number of numerical goals set**

c. Work as one team. (Sink differences)

d. Senior Management must push all other points constantly and consistently

284. Cost of Quality (COQ) includes

b) Preventive, detective & control

a) Preventive, appraisal & failure

c) Preventive, Corrective & Control

285. Quality without timeliness and cost consideration is_____

1. Meaningless

2. Beneficial

3. One of product deficiencies

4. Cheaper

286. Quality is ----------'s responsibility.

a. Nobody
b. **Everybody**
c. Somebody
d. Customer

287. AQL means –

a) Allowable Quality Level

b) Allocated Quality Level

c) Acceptable Quality Level

d) Allowed Quality Level

288. The concept that is an integral part of information technology function is –

a) Meeting the requirements

b) Producing a quality product

c) **Customer's view of quality is the correct view quality.**

289. Six sigma quality is -

a) 1.2 Defects per million lines of code

b) **3.4 Defects per million lines of code**

c) 6.1 Defects per million lines of code

290. Middle Level Management is _____ and _____

a. Visionary and tactical

b. **Visionary and directional**

c. Directional and tactical

291. Quality assurance is a _____ function.

a) line

b) staff

c) line & staff

292. Manufacturing Companies include

a) Agriculture

b) Mining

c) Construction

d) All the above

293. The total maximum points for MBNQA criteria is

a) 100 points

b) 500 points

c) 1000 points

d) 10000 points

294. Leadership category in MBNQA includes

a) Leadership system

b) Societal Responsibilities

c) Community Involvement

d) a,b & c

e) a & b only.

295. Organization at initial level can improve their performance by instituting

a) Project Management

b) Management Oversight

c) Quality Assurance

d) Change Control

e) All the above

f) Only a,b, & d

296. The number and types of measures for quality results depend upon factors such as

a the company's size

b types of products and services

c competitive environment

d all the above

297. The tactics which should be incorporated while giving constructive criticism are

A) Do it privately

B) Have the facts

C) Be prepared to help the worker to improve his/her performance

D) Be specific on expectation

E) Follow a specific process in giving criticism

F) A B C & E

G) All the above

298. People don't listen because they are

A) very impatient

B) too busy rehearsing what they will say next

C) self conscious

D) external stimuli

E) lack of motivation

F) all the above

299. The ability to apply beliefs, standards, guidelines, policies, procedures and values to a decision in a specific set of circumstances is known as

A) Motivation b) Recognition c) Judgment d) Time Management.

300. The complaint resolution processes are

a. get on your customers wave length

b. get the facts

c. establish and initiate an action program

d. Follow up with your customer

e. **all the above**

301. JIT was developed by

a) Dr. W. Edward deming

b) Dr. Leonard Zunin

c) Taiicho Ohno

d) Malcom Baldrige

302. Pareto concept was developed by –

a. Taiichi ohno

b. Edward Deming

c. Malcolm Baldrige

d. Joseph Juran

e. Pareto

303. What are the interpretations of 80 - 20 rule/pareto analysis?

a. 20% Of organisation's customers account for 80% of its revenue.

b. 20% Of organisation's products produce 80% of profits.

c. 3.80% of customers' complaints by volume are 20% of problems by categories.

304. What are the types of functions in FP?

A)External input type

B)external output type

C)logical internal file type

D)external interface file type

E)external inquiry type

305. What are the ways in which QA analyst can estimate frequency/severity of events?

1.Judgemental sampling. **2. Statistical sampling.**

306. Passwords should be changed

a. on regular basis

b. When you forget the password

c. when you leave the workplace

d. all of the above

e. none of the above

307. Which of the following techniques is most effective in encouraging participation of attendees in a meeting:

a. Role playing & requesting solutions

b. having the moderator to play the devil's role

c. requiring participation

308. Methods used to determine whether or not the process is under control and is helpful in identifying root cause of process problems which are causing defects

a. Prioritization matrix

b. **statistical process control**

c. pareto chart

d. control charts

309. The concept that is an integral part of information technology function is –

a) Meeting the requirements

b) Producing a quality product

c) Customer's view of quality is the correct view quality.

310. The Juran Trilogy refers to:

a) Quality Management, Quality Coordination, Continuous improvement

b) Quality planning, Quality Control, Quality improvement

c) Qualitative Management, Quantitative Management, Continuous improvement

d) Quality Assurance, Quality Control, Quality Audits

311. MBNQA Award criteria does not include

a) Leadership

b) **Technology**

c) Human Resources

d) Process Management

312. The three evaluation dimensions of Malcolm Baldrige National Quality Award are

a) Leadership, Strategic Planning, Business Results

b) Approach, Deployment, Results

c) Strategic Planning, Process Management, Business Results

d) Planning, Implementation, Feedback

313. Developing and asking questions for the purpose of collecting oral data to be used in an analysis or evaluation is called as

a) Negotiation

b) Facilitation

c) Interviewing

d) Effective listening

314. The tactics, which should be incorporated while giving constructive criticism, are

a) Do it privately

b) Have the facts

c) Be prepared to help the worker to improve his/her performance

d) Be specific on expectation

e) Follow a specific process in giving criticism

f) A B C & E

g) all the above

315. The ability to apply beliefs, standards, guidelines, policies, procedures and values to a decision in a specific set of circumstances is known as

a) Motivation

b) Recognition

c) Judgment

d) Time Management.

316. A service level agreement becomes a contract when the provider of service and customer are not in the same organization.

a) True

b) False

317. If the variation due to common causes results in a process operating outside of the customer specifications, the process is called

a) Incapable

b) Noncapable

c) Meaningless

d) Useless

318. Which of the following is not a description of the charter of quality function?

a) the person who is nominated by quality department to prepare quality charts

b) management's authorization to make quality happen

c) a job description for the quality function

d) explicitly explains to all the affected parties the function's scope and responsibilities.

319. According to SEI of CMU, to move from Maturity level 1 to level 3

a) it takes atleast 1 year

b) it takes atleast 3 years

c) it can be atmost 5 years

d) it depends on the Assessment process and Auditor

320. A successful Leader should always be open and honest

a) True

b) False

c) It depends

d) Need not be always

321. Function point analysis requires information on hardware and software for the application system.

a) True

b) False

322. _____ is a visual tool used to determine and understand the forces that drive and restrain a change.

a) Process Map

b) Cause and Effect Diagram

c) Force Field analysis

d) Matrix

323. Acceptable Quality Level Is To Accept Some Level Of Defects As Normal And Acceptable, As Many Believe That Defect Free Information Technology Products Are Not Practical Or Economical.

a) False

b) True

324. Internal control encompasses the plan of organization and all of the methods and measures adopted within a business to:

a) Safeguard its assets

b) Promote operational efficiency

c) Check the accuracy and reliability of accounting data

d) Encourage adherence to prescribed policies

e) All of the above

f) None of the above

325. To reduce risk associated with introducing changes to an application, _____ testing should be conducted during all stages of testing after a functional improvement or repair has been made.

a) Unit

b) System

c) Regression

d) Thread

326. Which of the following task is not performed by V&V management

a) Create the software V&V plan

b) Conduct Management Review of V & V

c) Support Management and Technical Reviews

d) Interface with Organizational and Supporting Processes

e) Conduct In-process reviews

327. Project team feels that an extra feature or a function is needed in the system but not included in the specifications. From quality point of view, this is considered ----------.

a. a correction
b. a failure
c. a change
d. an enhancement
e. **a defect**

328. Our products must be corrected so that they will eventually meet our customers' true quality needs.

T/F

327. We can produce products at our convenience and at any cost

T/**F**

328. Those responsible for quality must include –

1. Cost consideration

2. Delivery at required time

3. At required quality

4. 1 & 2

5. 2 & 3

6. All the above

329. Management must create an _____ a_____for quality to happen.

1. Environment , enthusiasm

2. Principles, reward

3. Empowerment, rewards

330. Management must make quality number one and then

reward quality work.

T/F

331. Monetary and non-monetary rewards can be considered by management to make people enjoy what they are doing.

T/F

332. Product inspections and testing can be eliminated by building processes to eliminate root causes of def and producing defect free products. **T**/F

333. The following comprise the processes - Which is correct

1.Policies, standards, procedures

2.standards, procedures, policies

3.Procedures, policies, standards

4.Policies, procedures, standards

334. Quality is not an attribute of product. F

335. Quality can be controlled only if it is measured. T

336. Judgment is an audit function T

337. Non conformance must be detected as early as possible - T/F

338 The concept that is an integral part of information technology function is –

1. Meeting the requirements

2. Producing a quality product

3. Customer's view of quality is the correct view quality.

339. The topics of standards and nonconformance should be discussed in every meeting to improve quality. T/F

340. Everyone should be involved in quality improvement programs. T/F

341. The responsibility of supervisor –

1. Teach subordinates how to perform

2. To evaluate subordinate performance

3. Assist and support workers until they become proficient

4. 1 &2

5. 1 & 3

6. 2 & 3

7. 1, 2 and 3

342. It is better to deal with more number of vendors. F

343. Base lining is quantitatively determining what you want to be F

344. Analysis is performed before base lining is done. F

347. Analysis is done to determine which area is in need of improvement. T

348. Vision provides an umbrella for focusing all the plans in order to achieve constancy of purpose. T

349. Six sigma quality is –

1. 1.2 Defects per million lines of code

2. 3.4 Defects per million lines of code

3. 6.1 Defects per million lines of code

350. The three levels of management are –

1. Lower, middle and upper

2. Executive, middle, line

351. Tools supporting processes to be ensured rather than processes supporting tools.

T/F

352. Quality control is the responsibility of people who evolves the processes. F

353. Quality defect is non conformance to standards. T

354. Verification activity is a tactic analysis. T

354. Common verification techniques are -

1.Walkthroughs, checkpoint reviews, inspections

2.Test plan, test execution

355. Match:

1. Strategic - a. Operational management

2. Tactical - b. Executive management

3. Operational - c. Middle management

1-b, 2-c, 3-a

356. Management must retain decision making authority to identify and resolve problems. T

357. Seven core processes are –

1.Baseline, analyse, vision and plan

2.Establish a TQM process

3.Install and improve processes

4.Control quality (walkthrough, reviews and inspections

5.Quality control (validation-test plan/test execution)

6.Perform quantitative analysis

7.Perform continuous improvement

358. Plan - Baseline, plan - approach part

Do - Set short term objectives , compare results, targets, revise approach if required

Check & act - check results, replan and recheck

Results - final results

Approach is repeatable.

359. Many technical personnel believe that standards inhibit their creativity. (**True**/False)

360. Achieving quality (defect-free products and services) is easy. (True/**False**)

361. The challenge of making quality happen is

a) a miniscule challenge

b) a monumental challenge

362. In Zero defect movement all responsibilities and defects are borne by the workers. (True/**False**)

363. The word "kick-off" is used in

a) Zero Defect Movement

b) Deming's Circle

c) QAI's Quality Improvement Model

364. Dr. W. Edwards Deming has stated that it takes _____ years to change a culture from an emphasis on productivity to an emphasis on quality.

(10 years/**20 years**/2 to 5 years/none of these)

365. Quality means meeting requirements. This is _____view.

(**producer's** /customer's)

366. Quality means fit for use. This is _____view.

(producer's /**customer's**)

367. Which one of the following definitions of quality is more important.

a)Quality means meeting requirements.

b)Quality means fit for use.

368. Quality control is a <u>line</u> function

369. Quality assurance is a <u>staff</u> function.

370. By what other two names is PDCA process referred to?

Deming's wheel, management cycle.

371. What does "Going Around the PDCA Circle" connote?

Continuous improvement

372. The no of principles in Dr. W. Edwards Deming's quality

principles is _____. (8/10/12/**14**/16)